25900023 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

NA SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBBT102 - FUNDAMENTALS OF ENZYMOLOGY AND RADIOBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer all questions

Each question carries **1** marks

1. What is polarity? [K] / [CO1]
2. What is the dilution factor? [K] / [CO2]
3. Define absolute specificity of an enzyme with an example. [K] / [CO3]
4. Write down Michael - Menten equation. [K] / [CO4]
5. What is a coenzyme? [K] / [CO3]
6. Define oxidoreductases with an example. [K] / [CO3]
7. Define isoenzymes [K] / [CO4]
8. How does temperature affect enzyme activity? [K] / [CO4]
9. How does Concentration of enzyme affect enzyme activity? [K] / [CO4]
10. Give any two properties of beta rays [K] / [CO6]

[1x10 = 10]

Part B

Answer any **4** questions

Each question carries **2** marks

11. What is the formula for calculating molality?	[K] / [CO2]
12. Give two properties of buffers.	[K] / [CO2]
13. What is the principle of scintillation counter?	[U] / [CO6]
14. How do activators affect enzyme activity?	[U] / [CO4]
15. Describe lyases with an example.	[K] / [CO3]
16. Explain uncompetitive inhibition	[U] / [CO5]

[2x4 = 8]

Part C

Answer any **3** questions

Each question carries **4** marks

17. Write notes on pH and pH scale	[U] / [CO2]
18. Explain the structure and working of pH meter.	[A] / [CO2]
19. Describe enzyme inhibition	[A] / [CO5]
20. Describe Geiger Muller counter	[A] / [CO6]
21. Elaborate on autoradiography	[A] / [CO6]

[4x3 = 12]

Part D

Answer any **2** questions

Each question carries **10** marks

22. Elaborate on acids and bases	[U] / [CO2]
23. Describe enzyme regulation.	[A] / [CO3]
24. Elaborate on the applications of various radioisotopes in biology	[E] / [CO6]
25. Describe the mechanism of enzyme action	[A] / [CO3, CO5]

[10x2 = 20]